

Graph Neural Network Assisted Deep Reinforcement Learning Based Hierarchical Routing in Mega LEO Satellite Constellation

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Abstract—Mega low-Earth-orbit (mLEO) constellations with inter-satellite links (ISLs) enable wide-area data delivery, but routing in such systems is challenged by large network scale, time-varying link availability, and spatially concentrated hotspot traffic. This paper studies task route setup over the satellite ISL segment and develops a learning-based hierarchical framework that decomposes source-to-egress ISL route setup into inter-region route setup and intra-region path construction. At the upper layer, regions perform synchronized neighbor message passing on the slot snapshot to generate local embeddings, and a parameter-sharing graph neural network (GNN)-assisted Proximal Policy Optimization (PPO) policy determines the inter-region route through sequential local decisions of the currently active region agent. This design avoids satellite-level real-time global search while preserving sensitivity to regional load and boundary-link conditions. To improve training under sparse success feedback, a heuristic teacher-guided warm start provides soft guidance in the early stage of policy learning. At the lower layer, a load-aware shortest-path method constructs executable intra-region segments, and each segment is revalidated at its actual start slot so that local repair can be triggered by link disruptions or capacity blocking. Using a simulation environment built from real two-line element (TLE) data and the Simplified General Perturbations 4 (SGP4) propagator, we evaluate the proposed method under different task sizes and runtime ISL failures. The results show that the proposed method achieves a favorable trade-off among task success rate, penalized average delay, and load balance, while maintaining robust performance under runtime ISL failures.

Index Terms—LEO constellation, route planning, graph neural network, deep reinforcement learning.

I. INTRODUCTION

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WITH the growth of the information industry and advances in reusable rocket technologies [1], mega low Earth orbit (mLEO) satellite constellations have developed rapidly in recent years. Such constellations for broadband access and global coverage are important components of future 6G networks [2]–[4]. Compared with traditional satellite communication systems, mLEO constellations offer lower propagation delay, better coverage continuity, and stronger redundancy [5]. Their large deployment scale also reduces average manufacturing and launch cost through mass production and multi-satellite launches [6]. More importantly, inter-satellite links (ISLs) enable multi-hop space networking, allowing data to be relayed directly across satellites and reducing dependence on ground facilities [7].

Despite these advantages, providing global end-to-end connectivity in mLEO constellations remains challenging. Limited onboard resources require congestion-aware load allocation under scarce capacity [8]. ISLs may become intermittently unavailable because of acquisition, pointing, and tracking (APT) errors [9] and other disturbances. In addition, orbital dynamics and spatially concentrated task arrivals can produce localized hotspot traffic that concentrates load on a small set of links, and shortest-path routing alone may therefore cause persistent hotspots and cascading congestion [10]. The key challenge is to maintain high reachability and low delay while distributing load under rapidly changing topology and intermittent link failures.

In recent years, routing for mLEO constellations has attracted increasing attention. Traditional methods such as the selective iterative Dijkstra algorithm (SIDA) were studied in [11], [12], but routing that depends on global information faces synchronization and dissemination difficulties at large scale. Learning-based routing has also been explored [13]–[15]. These methods usually make decisions from local observations, but many do not explicitly model localized hotspot traffic. Another line of work combines graph neural networks (GNNs) with deep reinforcement learning (DRL) [16], [17] to extract broader structural features. However, the evaluated constellations are still limited in size, typically around 1,000 satellites or fewer, and explicit load-balancing design for localized hotspot traffic remains insufficient.

To complement existing studies, we consider hotspot-aware task routing in an mLEO constellation with thousands of satellites and directional backhaul traffic under localized hotspot traffic. The considered task route starts from a source satel-

lite and ends at a gateway-visible egress satellite. The S2G segment is represented by the selected egress satellite and is not separately optimized. We develop a learning-based hierarchical routing framework that decomposes source-to-egress ISL route setup into inter-region route setup with synchronized region-level embedding preparation and intra-region path construction. At the beginning of each slot, each region builds local features from the current snapshot and exchanges compact regional and boundary-link messages only with adjacent regions for several synchronous rounds to obtain slot-level local embeddings. Based on these embeddings, a parameter-sharing regional policy is invoked sequentially by the currently active region agent to determine the next region hop. The inter-region route is thereby established through local decisions. We formulate routing as a joint optimization of source-to-egress ISL delay and congestion-aware load balance. For upper-layer route setup, we combine a GNN encoder with Proximal Policy Optimization (PPO) and introduce a heuristic teacher-guided warm start [18] in training, where a computable reference policy provides soft guidance in early episodes to ease the cold-start problem caused by sparse success feedback at large scale. The main contributions of this paper are summarized as follows:

- 1) We propose a scalable learning-based hierarchical routing framework for mLEO constellations that decomposes source-to-egress ISL route setup into inter-region route setup with synchronized region-level embedding preparation and intra-region path construction. The upper layer performs a finite number of synchronized local-neighbor message-passing rounds on the region graph to generate slot-level regional embeddings, and a parameter-sharing regional policy establishes the inter-region route through sequential local decisions of the currently active region agent. At the lower layer, executable inter-satellite path segments are constructed inside each visited region, with execution-time revalidation and local repair when necessary.
- 2) We formulate a joint optimization problem of source-to-egress ISL delay and network load balance under localized hotspot traffic. Based on slot-level regional embeddings, we use a GNN-assisted PPO scheme for inter-region route setup, in which each regional agent uses only its local observation and candidate-direction information. We further introduce a heuristic teacher-guided warm start to improve early-stage training stability and convergence speed.
- 3) We construct an mLEO simulation environment using real two-line element (TLE) [19] data and the Simplified General Perturbations 4 (SGP4) propagator tool. The environment models satellite orbits, link visibility, gateway station connectivity, time-indexed link reservation, and runtime ISL failures, enabling realistic network topology and visibility conditions for routing evaluation. Experimental results across multiple scenarios show that the proposed method achieves a favorable trade-off among multiple performance metrics compared with representative benchmark algorithms.

The remainder of this article is organized as follows. Section II introduces related work. Section III presents the system model and problem formulation. Section IV describes the proposed method. The simulation results are shown and discussed in Section V. Finally, Section VI concludes this article.

II. RELATED WORK

A. Hierarchical Routing Methods

In hierarchical routing, [20] proposed LoHi, a load-aware hierarchical information-centric routing protocol for large-scale LEO satellite networks. It partitions the constellation into satellite groups, uses logical path identifiers to represent adjacent-group connectivity, and adjusts forwarding according to group-level path load and real-time ISL conditions. This design improves scalability and load awareness, but it mainly operates as a protocol and forwarding architecture based on hierarchical tables and heuristic decisions. [21] proposed a quality-of-service (QoS)-oriented hierarchical scheme in which a double deep Q-network (DDQN) selects links to balance path length and congestion. However, the evaluation was conducted on relatively small networks, and the authors also noted potential uncertainty in convergence. [22] decomposed routing into control-domain-partition-driven distributed routing and link-state-aware dynamic decisions, restricting link-state exchange to local and neighboring domains and using a deep Q-network (DQN) for path selection. [23] proposed a software-defined networking (SDN)-based clustered routing scheme in which controllers handle intra-cluster state collection and rule installation, while inter-cluster forwarding adopts destination-geography heuristics to reduce signaling overhead. Without global network state, however, purely distributed inter-cluster routing may produce suboptimal paths, including zig-zag routes.

These studies show that partition-based routing can reduce signaling overhead and improve scalability, but they are mainly based on protocol mechanisms, heuristic forwarding, or per-satellite decisions. In this paper, we focus on hotspot-aware source-to-egress ISL route setup in a larger mLEO constellation and study learned inter-region route setup with GNN-extracted local embeddings, PPO-based directional action selection, and executable intra-region path construction under time-varying links.

B. Intelligent Routing Based on GNN and DRL

Machine learning, especially DRL, has attracted increasing attention in satellite network routing, and many recent methods therefore combine DRL with GNNs. [24] proposed the GRLR method, which uses a GNN to extract graph features and applies DRL to learn the path-selection policy. This method can handle dynamic topology changes and time-varying link characteristics, thereby reducing end-to-end delay and improving network throughput. [25] proposed an inter-satellite routing method that integrates GNNs with DRL. Its key idea is to use a graph convolutional network (GCN) to capture topology dynamics and output node feature vectors,

while DRL optimizes routing decisions. [26] proposed a DRL-based routing mechanism using a spatiotemporal GNN (ST-GNN). It introduces routing-associated age of information (RAoI) as a timeliness metric and designs a spatiotemporal feature extraction module. [27] proposed a fully distributed dynamic packet-routing algorithm based on GNN-enhanced multi-agent reinforcement learning (GNN-MADRL). It uses a graph attention network (GAT) to process local topology and generate node embeddings, together with multi-agent coordination.

These studies present several GNN-assisted DRL-based routing methods. Some also evaluate mLEO constellations with more than one thousand satellites. However, most of them focus on general dynamic routing or timeliness optimization rather than hotspot-aware load balancing under spatially localized traffic concentration. In addition, route planning over very large constellations still faces a scale challenge, because satellite-level planning becomes difficult as the constellation grows. The present work therefore focuses on hierarchical source-to-egress ISL route setup with GNN-assisted region-level representation learning and synchronized regional decisions for large-scale mLEO constellations under localized hotspot traffic.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section first introduces the system model of the mLEO satellite network. The source-to-egress ISL delay and resource occupancy of tasks are then formulated. Finally, we present the joint optimization problem of source-to-egress ISL delay and load balance. The main notation is summarized in Table I.

A. Constellation Model

We consider an mLEO satellite constellation with thousands of satellites interconnected by ISLs, where traffic can be downlinked through a gateway station. The satellite set is denoted by $\mathcal{V} = \{1, 2, \dots, N\}$, where N is the constellation size. The gateway station is denoted by g and has a fixed geographic location. We adopt the Walker-delta constellation specified by $i/N_L/M_L/F$ in [28], where i , N_L , M_L , and F denote the orbital inclination, the number of orbital planes, the number of satellites per plane, and the phasing factor, respectively.

For inter-satellite connectivity, a fixed candidate ISL neighbor set $\mathcal{N}_i^{\text{cand}}$ is predefined for each satellite i . Owing to the structural properties of Walker-delta constellations, each satellite can establish ISLs with its two adjacent satellites in the same orbital plane and with adjacent satellites in the two neighboring planes. A wrap-around rule is adopted to preserve connectivity at the boundaries. When the orbital-plane index or phase index exceeds its valid range, the candidate connection is formed with the satellite on the opposite boundary. We define the ISL edge set as $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. ISLs are assumed to be bidirectional under normal conditions, that is, if $(i, j) \in \mathcal{E}$, then $(j, i) \in \mathcal{E}$. This connectivity model keeps the routing action space structurally consistent in large-scale networks.

For satellite-to-ground (S2G) communication, connectivity between the gateway and satellites is determined by S2G

TABLE I: NOTATIONS

Notation	Definition
$\mathcal{V}$	Satellite set
N	Number of satellites
g	Gateway station
$\mathcal{N}_i^{\text{cand}}$	Candidate ISL neighbor set of satellite i
$\mathcal{E}$	Candidate ISL set
$\theta_{\min}$	Minimum elevation angle threshold
$\mathcal{V}_t^g$	Gateway-visible satellites at time t
$\mathcal{T}$	Time-slot index set
T_{sim}	Number of simulation slots
Δt	System time slot length
$\mathbf{r}_i(t)$	Position vector of satellite i at time t
$\mathbf{r}_g$	Position vector of gateway station g
$\mathcal{K}$	Task set
k	Task index
t_k	Generation slot of task k
$t_{k,h}$	Start time of hop h of task k
C	ISL capacity
$\rho_{ij}^{\text{bg}}(t)$	Background occupancy ratio
$\rho_{ij}^{\text{bg,max}}$	Upper bound of the background occupancy ratio
$\rho_{ij}(t)$	Total occupancy ratio of link (i, j) at time t
ρ_{blk}	Blocking threshold for link admission
$\alpha_{ij}^k(t)$	Reserved occupancy ratio of task k on link (i, j) at time t
α_0	Fixed occupancy ratio per admitted task
$a_{ij}^{\text{geo}}(t)$	Geometric availability indicator
$a_{ij}^{\text{blk}}(t)$	Blocking availability indicator
$a_{ij}^{\text{ot}}(t)$	Runtime-failure availability indicator
$a_{ij}(t)$	Link availability indicator
c_0	Speed of light
$\mathcal{E}(t)$	Available link set at time slot t
s_k	Source node of task k
d_k	Gateway-visible egress satellite of task k
B_k	Data size of task k
P_k	Source-to-egress ISL path of task k
H_k	Hop count of task k
$H_{\text{max}}^{\text{max}}$	Maximum satellite hop count
$\tau_{ij}^{\text{prop}}(t)$	Propagation delay
$\tau_{k,h}^{\text{tx}}$	Transmission delay of task k at hop h
T_k	Source-to-egress ISL delay of task k
T_{pen}	Penalty delay assigned to failed tasks
$\bar{T}_k$	Penalized delay of task k
$\mathcal{V}^{\text{reg}}$	Region set
$\mathcal{E}^{\text{reg}}$	Region-edge set
$\kappa(\cdot)$	Satellite-to-region mapping
P_h^{seg}	Satellite-level segment at inter-region step h
$\text{JFI}(t)$	Network-wide Jain's fairness index
$\bar{T}$	Long-term average penalized delay
$\bar{J}$	Long-term average Jain's fairness index
λ_T	Weight coefficient of the delay term
λ_J	Weight coefficient of the load-balance term

visibility and elevation angle. We consider a single gateway station with a fixed geodetic location to support S2G traffic transmission. The gateway employs a minimum elevation angle threshold $\theta_{\min}$. If the LOS condition is satisfied and the elevation angle is no smaller than $\theta_{\min}$, the satellite is regarded as visible to the gateway and an S2G link can be established. The set of satellites visible to the gateway at time t is denoted by $\mathcal{V}_t^g$. We use real TLE data to describe satellite orbits and apply the SGP4 orbit propagator over the entire simulation period to compute the time-varying position of each satellite. This setting captures realistic geometric relationships and visibility variations. The system model is shown in Fig. 1.

We assume that the system operates in discrete time slots, following our previous work [29]. The slot index set is denoted by $\mathcal{T} = \{0, 1, \dots, T_{\text{sim}} - 1\}$, and the duration of each slot is

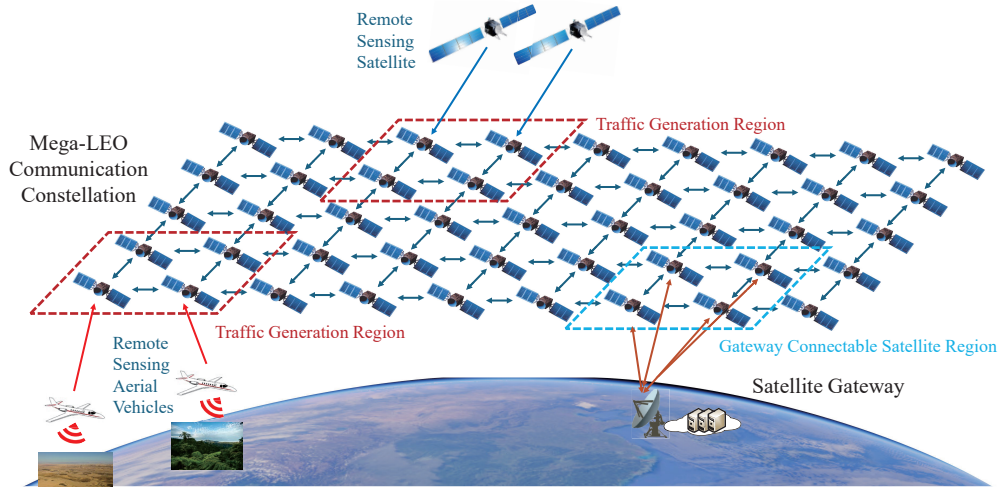


Fig. 1: The network model of the proposed mLEO constellation.

Δt . At each slot, SGP4 provides the satellite position vector $\mathbf{r}_i(t) \in \mathbb{R}^3$, while the gateway position vector is denoted by $\mathbf{r}_g \in \mathbb{R}^3$. ISL availability and S2G link availability are assumed to remain unchanged within one slot. Based on link visibility and geometric distance, link availability and propagation delay can therefore be determined.

B. Communication and Delay Model

We model traffic at the task level. The task set is denoted by $\mathcal{K}$. For any task $k \in \mathcal{K}$, its data volume is set to be $B_k > 0$. In each time slot, the system generates one task for scheduling. Localized burstiness is modeled as spatial concentration over a refresh window, where consecutive tasks are sampled from localized source and egress patches.

We assume a constant inter-satellite transmission capacity for ISLs, which is supported by in-orbit experiments in [30]. Given the relative stability of optical ISLs and the predictable orbit geometry, this assumption is suitable for routing in most cases. For any ISL (i, j) , the nominal transmission capacity is set to C , and all ISLs are assumed to have the same capacity.

To better reflect practical operation, each ISL carries a small amount of background traffic for necessary inter-satellite information exchange. This traffic occupies part of the total link capacity. The background occupancy ratio on link (i, j) at time t is denoted by $\rho_{ij}^{\text{bg}}(t) \in [0, 1]$ and satisfies

$$0 \leq \rho_{ij}^{\text{bg}}(t) \leq \rho_{\text{bg},\max} < 1, \quad (1)$$

where $\rho_{\text{bg},\max}$ is the upper bound of the background occupancy ratio.

We further assume that ISLs employ mature multiplexing techniques, and multiple tasks can be transmitted in parallel on one link. To limit decision dimensionality and system complexity, a reservation-based resource-allocation mechanism is adopted as in [31]: once a task is scheduled, its allocated communication resources are reserved and remain occupied until completion. Resource allocation is characterized by a capacity occupancy ratio. Let $\alpha_{ij}^k(t) \in [0, 1]$ denote the

fraction of C reserved for task k on link (i, j) at time t . The total occupancy ratio of link (i, j) is defined as

$$\rho_{ij}(t) = \rho_{ij}^{\text{bg}}(t) + \sum_{k \in \mathcal{K}_{ij}(t)} \alpha_{ij}^k(t), \quad (2)$$

where $\mathcal{K}_{ij}(t)$ denotes the set of tasks that are either transmitting on link (i, j) at time t or have completed resource reservation but have not yet started transmission. For simplicity, we assume that each task admitted to a link uses a fixed occupancy ratio $\alpha_0 \in (0, 1)$ throughout its reservation and transmission period. Such reservations are time-indexed in simulation, so $\rho_{ij}(t)$ includes active transmissions and active reservations with release times later than t . The number of tasks that can be carried in parallel is therefore limited by the total occupancy constraint.

We consider that link availability is determined by three factors: geometric availability, occupancy-threshold blocking, and runtime failures. We use $a_{ij}^{\text{geo}}(t) \in \{0, 1\}$ to denote the geometric-availability indicator determined by the line-of-sight (LOS) condition. Since practical systems typically cannot operate at the theoretical capacity limit, an occupancy threshold $\rho_{\text{blk}} \in (0, 1]$ is introduced to model blocking caused by link occupancy. To avoid timing ambiguity between admission checking and occupancy updates, the link occupancy ratio immediately before admitting new tasks at time t is denoted by $\rho_{ij}(t^-)$. A link is regarded as unavailable for routing and scheduling once the occupancy reaches the threshold:

$$a_{ij}^{\text{blk}}(t) = \mathbb{I}\{\rho_{ij}(t^-) < \rho_{\text{blk}}\}. \quad (3)$$

To capture disruptions caused by non-geometric factors such as transient hardware glitches, APT errors, and short-term anomalies, we introduce a binary runtime failure indicator $a_{ij}^{\text{otg}}(t) \in \{0, 1\}$. The stable-link setting is obtained by setting $a_{ij}^{\text{otg}}(t) \equiv 1$ for all links and slots. In the runtime-failure setting, failures are generated by a scheduled link-disruption process. Let P_{lf} denote the trigger period, p_{lf} the per-link failure probability at a trigger slot, and D_{lf} the failure duration in slots. At every trigger slot τ satisfying

$(\tau - 1) \bmod P_f = 0$, each candidate ISL that is not already in a failure interval is independently checked. Each checked ISL is determined to suffer a runtime failure with probability p_{lf} . These failure events are independent and identically distributed across checked ISLs at each trigger slot. If link (i, j) is determined to fail at slot τ , both directions of this ISL are set to unavailable during $[\tau, \tau + D_{lf})$ and are restored after the interval. Equivalently, $a_{ij}^{\text{otg}}(t) = 0$ when t falls into an active runtime-failure interval of link (i, j) , and $a_{ij}^{\text{otg}}(t) = 1$ otherwise. The final availability of link (i, j) at time t is defined as

$$a_{ij}(t) = a_{ij}^{\text{geo}}(t) \cdot a_{ij}^{\text{blk}}(t) \cdot a_{ij}^{\text{otg}}(t). \quad (4)$$

When $a_{ij}(t) = 0$, link (i, j) is unavailable at time t . Based on the above model, the delay of each hop in multi-hop transmission can be derived. The routing path of task k is an ordered node sequence $P_k = (v_{k,0}, v_{k,1}, \dots, v_{k,H_k})$, where H_k is the hop count. A store-and-forward mode is assumed, so the task can enter the next hop only after completing transmission on the current hop. Propagation delay is computed from geometric distance. If task k uses link (i, j) at time t , the propagation delay is

$$\tau_{ij}^{\text{prop}}(t) = \frac{\|\mathbf{r}_i(t) - \mathbf{r}_j(t)\|}{c_0}, \quad (5)$$

where $\mathbf{r}_i(t)$ is the satellite position vector and c_0 is the speed of light. Transmission delay depends on the task data volume and the allocated throughput. During execution, each hop is admitted only if the link is available and sufficient occupancy headroom exists at the actual transmission-start time, that is,

$$a_{ij}(t) = 1, \quad \rho_{ij}(t^-) + \alpha_0 \leq \rho_{\text{blk}}, \quad (6)$$

where t^- denotes the pre-admission state right before resources are allocated to the new task at time t . Once admitted, the task transmits on that link with occupancy ratio α_0 until completion. The transmission delay of hop h is defined as

$$\tau_{k,h}^{\text{tx}} = \frac{B_k}{\alpha_0 \cdot C}, \quad (7)$$

where $t_{k,h}$ denotes the start time when task k enters the transmission stage of hop h , which is obtained recursively from the completion time of the preceding hop by

$$t_{k,h+1} = t_{k,h} + \tau_{v_{k,h}, v_{k,h+1}}^{\text{prop}}(t_{k,h}) + \tau_{k,h}^{\text{tx}}, \quad h = 0, \dots, H_k - 1. \quad (8)$$

The source-to-egress ISL delay of task k is the sum over hops:

$$T_k = \sum_{h=0}^{H_k-1} \left(\tau_{v_{k,h}, v_{k,h+1}}^{\text{prop}}(t_{k,h}) + \tau_{k,h}^{\text{tx}} \right). \quad (9)$$

C. Problem Formulation

Based on the communication and delay model in Section III-B, we formulate a long-term routing optimization problem. For any task $k \in \mathcal{K}$, the task is generated at slot t_k at source

node $s_k \in \mathcal{V}$, and its destination node is the gateway-visible egress satellite $d_k \in \mathcal{V}$. The route therefore starts from s_k and ends at d_k . The S2G segment is represented by the choice of d_k and is not optimized as a separate routing segment. The network is modeled as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the satellite set and $\mathcal{E}$ is the candidate ISL set. The availability of link $(i, j) \in \mathcal{E}$ at time t is described by $a_{ij}(t) \in \{0, 1\}$, and the set of available links is defined as

$$\mathcal{E}(t) \triangleq \{(i, j) \in \mathcal{E} \mid a_{ij}(t) = 1\}. \quad (10)$$

For task k , its routing path is represented by an ordered node sequence

$$P_k = (v_{k,0} = s_k, v_{k,1}, \dots, v_{k,H_k} = d_k), \quad (11)$$

where H_k is the hop count. The reachability requirement of each hop is

$$(v_{k,h}, v_{k,h+1}) \in \mathcal{E}(t_{k,h}), \quad \forall k, \quad \forall h = 0, 1, \dots, H_k - 1. \quad (12)$$

According to Section III-B, the total occupancy ratio of link $(i, j) \in \mathcal{E}$ at time t is $\rho_{ij}(t)$ and it is subject to the occupancy threshold ρ_{blk} . A task must satisfy the admission condition in (6) when it enters the transmission stage of a link.

For the delay metric, the propagation delay of each hop $\tau_{ij}^{\text{prop}}(t)$, the per-hop transmission delay $\tau_{k,h}^{\text{tx}}$, and the source-to-egress delay T_k follow (5), (7), and (9), respectively. To handle failed tasks within a unified objective, we introduce a penalty delay constant $T_{\text{pen}} > 0$ and define the penalized delay $\tilde{T}_k$ as

$$\tilde{T}_k = \begin{cases} T_k, & \text{if task } k \text{ successfully reaches } d_k, \\ T_{\text{pen}}, & \text{otherwise.} \end{cases} \quad (13)$$

Failure events include, but are not limited to, the following cases: the satellite link is unavailable, the admission condition is violated, or the destination is not reached before the maximum satellite hop limit. The long-term average penalized delay is defined as

$$\bar{T} \triangleq \mathbb{E}[\tilde{T}_k]. \quad (14)$$

We use the global Jain's fairness index (JFI) [32] to characterize network load balance, where the statistics are computed over the currently available ISL set $\mathcal{E}(t)$. Since $\rho_{ij}(t)$ already represents the link occupancy ratio, the JFI at time t is given by

$$\text{JFI}(t) = \begin{cases} \frac{\left(\sum_{(i,j) \in \mathcal{E}(t)} \rho_{ij}(t) \right)^2}{|\mathcal{E}(t)| \sum_{(i,j) \in \mathcal{E}(t)} \rho_{ij}^2(t)}, & \sum_{(i,j) \in \mathcal{E}(t)} \rho_{ij}(t) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (15)$$

When no link carries traffic, such as under total link failure, we set $\text{JFI}(t) = 0$ as a sentinel value to explicitly indicate this abnormal case. Since JFI is a network-wide fairness metric, it

is later complemented by HSI in Section V-C to capture load concentration on heavily utilized links. To reflect long-term balance, we define the time-average JFI as

$$\bar{J} \triangleq \mathbb{E}[\text{JFI}(t)]. \quad (16)$$

In summary, under long-term task arrivals, we seek a routing policy π that jointly optimizes the average penalized delay and the long-term load balance subject to time-varying link availability and capacity-admission constraints. We adopt a weighted-sum single-objective formulation:

$$\max_{\pi} -\lambda_T \bar{T} + \lambda_J \bar{J} \quad (17)$$

where $\lambda_T > 0$ and $\lambda_J > 0$ are weight coefficients. Here, $\bar{T}$ captures task-level source-to-egress transmission efficiency and incorporates the cost of failure through T_{pen} , while $\bar{J}$ regularizes network-level load distribution to avoid persistent overload on hotspot links caused by shortest-path-only routing. The proposed optimization problem is described as:

$$\max_{\pi} -\lambda_T \mathbb{E}[\tilde{T}_k] + \lambda_J \mathbb{E}[\text{JFI}(t)] \quad (18)$$

$$\text{s.t. } (v_{k,h}, v_{k,h+1}) \in \mathcal{E}(t_{k,h}), \forall k, \forall h, \quad (18a)$$

$$\rho_{v_{k,h} v_{k,h+1}}(t_{k,h}^-) + \alpha_0 \leq \rho_{\text{blk}}, \forall k, \forall h, \quad (18b)$$

$$\rho_{ij}(t) \leq \rho_{\text{blk}}, \forall (i, j) \in \mathcal{E}, \forall t, \quad (18c)$$

$$v_{k,h} \neq v_{k,h'}, \forall k, \forall h \neq h', \quad (18d)$$

$$H_k \leq H_{\text{sat}}^{\max}, \forall k. \quad (18e)$$

Constraint (18a) enforces physical reachability at the corresponding start time of each hop. Constraint (18b) requires each hop to satisfy the admission condition under the pre-admission occupancy convention. Constraint (18c) limits the total occupancy of every link at any time. Constraint (18d) prevents repeated nodes in a path and thus avoids loops and backtracking. Constraint (18e) bounds the hop count by the maximum satellite hop limit $H_{\text{sat}}^{\max}$.

Since routing, admission, and path-structure constraints involve discrete decisions coupled with varying link availability and network load, the above formulation can be viewed as a long-term mixed-integer nonlinear optimization problem. Conventional optimization methods often struggle to obtain high-quality solutions at the target constellation scale under such dynamics. We therefore adopt a GNN-assisted PPO approach, in which region-level graph embeddings support sequential inter-region decisions for hotspot-aware task routing.

IV. PROPOSED METHOD

This section presents the proposed hierarchical routing framework for mLEO constellations. We first describe the overall routing mechanism, then introduce the upper-layer inter-region route-setup design, including GNN-based feature extraction and PPO-based directional action selection. We also present the warm-start mechanism used in training. Finally, we describe the intra-region path construction strategy and analyze the overall complexity.

A. Hierarchical Routing Mechanism

In mLEO satellite networks, source-to-egress ISL route setup is difficult because of the large network scale and time-varying link conditions. Collecting and synchronizing global network state over thousands of nodes is costly for online decision making, whereas purely hop-by-hop local decisions tend to accumulate errors on long paths. In addition, ISLs may fail intermittently because of APT errors, and a single intermediate failure can invalidate the whole route. To address these issues, we decompose routing into inter-region route setup and intra-region path construction.

We partition the constellation into N_{reg} regions, denoted by $\mathcal{V}^{\text{reg}} = \{1, 2, \dots, N_{\text{reg}}\}$, and use a mapping $\kappa : \mathcal{V} \rightarrow \mathcal{V}^{\text{reg}}$ to assign each satellite to a region. Regions are grouped by orbital plane and phase indices and remain fixed during the simulation. Region adjacency follows a four-neighbor grid (up/down/left/right) with wrap-around at boundaries, which yields the inter-region topology $\mathcal{G}^{\text{reg}} = (\mathcal{V}^{\text{reg}}, \mathcal{E}^{\text{reg}})$.

For task k with source s_k and destination d_k , we map them to regions $c_k^{\text{src}} = \kappa(s_k)$ and $c_k^{\text{dst}} = \kappa(d_k)$. We then perform learning-based inter-region route setup on the region graph after synchronized local-neighbor embedding preparation. At planning step h , using the slot-level local embeddings generated by a finite number of synchronized local-neighbor message-passing layers, the agent of the current region c_h selects the next region c_{h+1} under a parameter-sharing policy using its local observation. The selected inter-region transition ($c_h \rightarrow c_{h+1}$) is first instantiated on the planning snapshot: the system determines the boundary pair and constructs the corresponding executable intra-region segment P_h^{seg} by a load-aware shortest-path algorithm. During execution, each segment is revalidated at its actual start slot using the latest link availability and time-indexed reservation state. The source-to-egress ISL route is obtained by concatenating all successfully instantiated satellite-level segments $\{P_h^{\text{seg}}\}$.

To improve execution robustness, we further introduce local repair. If a planned segment becomes infeasible because of link disruption or capacity blocking, the lower layer reruns intra-region path construction once with the same entry and exit satellites. If the blocked part is a boundary ISL, the boundary pair is reselected within the same inter-region direction. If repair fails, the task is treated as failed for that segment. In this case, the next intra-region segment is replanned accordingly.

The proposed hierarchical routing method is illustrated in Fig. 2. Overall, region partitioning reduces the decision scale and yields a two-stage routing design: inter-region route setup keeps global planning complexity manageable, while intra-region path construction ensures path executability and enables rapid local recovery. Since inter-region route setup involves aggregating high-dimensional load and link availability information over a nontrivial region graph, we use GNN-based representation learning and DRL-based decision making at the region level, while adopting a traffic-aware shortest-path method for intra-region path construction because the local topology is much smaller.

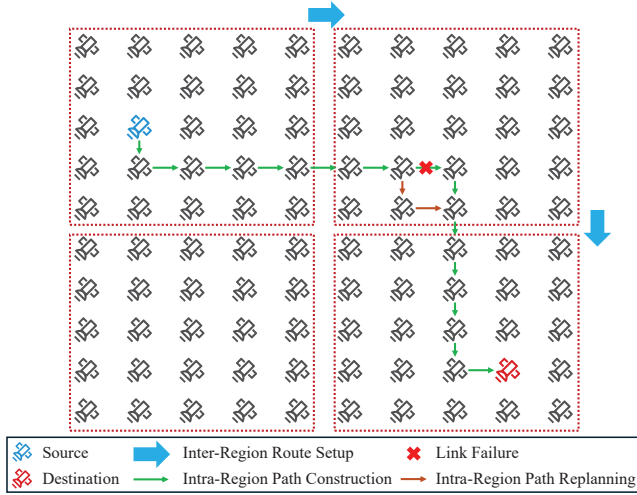


Fig. 2: The proposed hierarchical routing procedure.

B. Graph Neural Network-Based Feature Extraction

At each planning step, the upper layer performs one inter-region directional action selection on the directed graph $\mathcal{G}^{\text{reg}} = (\mathcal{V}^{\text{reg}}, \mathcal{E}^{\text{reg}})$, where $\mathcal{V}^{\text{reg}}$ is the region set and $\mathcal{E}^{\text{reg}} \subseteq \mathcal{V}^{\text{reg}} \times \mathcal{V}^{\text{reg}}$ contains directed edges between adjacent regions. For each region $c \in \mathcal{V}^{\text{reg}}$, its adjacency is indexed by the four-direction set $\mathcal{D}$, and $\mathcal{N}_d(c)$ denotes the neighboring region reachable from c along direction $d \in \mathcal{D}$.

At slot t , each region c is described by the node feature

$$\mathbf{x}_c(t) = \begin{bmatrix} \rho_c^{\text{in}, p90}, \bar{\rho}_c^{\text{in}}, \rho_c^{\text{in}, \text{std}}, \\ \rho_c^{\text{bun}, p90}, \bar{\rho}_c^{\text{bun}}, \rho_c^{\text{bun}, \text{std}} \end{bmatrix}. \quad (19)$$

where $\rho_c^{\text{in}, p90}$, $\bar{\rho}_c^{\text{in}}$, and $\rho_c^{\text{in}, \text{std}}$ are the 90th percentile, mean, and standard deviation of ISL utilization over internal ISLs of region c , while $\rho_c^{\text{bun}, p90}$, $\bar{\rho}_c^{\text{bun}}$, and $\rho_c^{\text{bun}, \text{std}}$ are computed over ISLs incident to boundary satellites of region c .

For each directed edge $(c \rightarrow c') \in \mathcal{E}^{\text{reg}}$, we define the edge feature

$$\mathbf{e}_{c \rightarrow c'}(t) = \begin{bmatrix} \rho_{c \rightarrow c'}^{p90}, \rho_{c \rightarrow c'}^{\min}, \rho_{c \rightarrow c'}^{\text{sum}}, \rho_{c \rightarrow c'}^{\text{std}} \end{bmatrix}. \quad (20)$$

where these terms denote the 90th percentile, minimum, sum, and standard deviation of the utilizations of boundary ISLs connecting region c and region c' , respectively.

Internal-region statistics summarize congestion inside a region, while direction-specific boundary-link statistics retain the load state of each candidate crossing direction. Directly feeding these handcrafted statistics to the policy network would expose only local summaries. To encode both regional context and boundary-link conditions, we use an edge-conditioned GraphSAGE encoder [33], [34]. The initial node embedding is

$$\mathbf{h}_c^{(0)} = f_{\text{in}}(\mathbf{x}_c), \quad (21)$$

where $f_{\text{in}}(\cdot)$ is a multilayer perceptron (MLP). We then perform L synchronized local-neighbor message-passing layers

on $\mathcal{G}^{\text{reg}}$. For any directed edge $(i \rightarrow j)$, the edge-conditioned message at layer l is computed by a message function $\phi_l(\cdot)$:

$$\mathbf{m}_{i \rightarrow j}^{(l)} = \phi_l \left(\left[\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \mathbf{e}_{i \rightarrow j} \right] \right), \quad (22)$$

where $[\cdot]$ denotes vector concatenation. For node j , messages from all incoming edges are aggregated as

$$\mathbf{a}_j^{(l)} = \sum_{i: (i \rightarrow j) \in \mathcal{E}^{\text{reg}}} \mathbf{m}_{i \rightarrow j}^{(l)}. \quad (23)$$

The next layer embedding is obtained by adding the self-loop term and applying an update network:

$$\mathbf{h}_j^{(l+1)} = \text{up}_l \left(\text{self}_l \left(\mathbf{h}_j^{(l)} \right) + \mathbf{a}_j^{(l)} \right), \quad (24)$$

where $\text{self}_l(\cdot)$ is the self-loop linear transform and $\text{up}_l(\cdot)$ is the update MLP. After L layers, the node embedding of region c is

$$\mathbf{z}_c = \mathbf{h}_c^{(L)}. \quad (25)$$

Node embeddings alone do not explicitly distinguish different outgoing directions from the same region. We therefore introduce an edge readout network to generate a direction-specific embedding for each directed region edge:

$$\mathbf{z}_{c \rightarrow c'}^e = \text{edge_ro} \left([\mathbf{z}_c, \mathbf{z}_{c'}, \mathbf{e}_{c \rightarrow c'}] \right). \quad (26)$$

The encoder thus outputs both the region-state representation set $\{\mathbf{z}_c\}$ and the action-specific representation set $\{\mathbf{z}_{c \rightarrow c'}^e\}$ for inter-region decision making. The encoder can learn structural topology-conditioned regional representations on the region graph. The node features summarize internal-region and boundary-link load states, while the directed edge features describe the load condition of candidate inter-region transitions. Unlike direct use of fixed engineered summaries as PPO inputs, the edge-conditioned message-passing process updates each region representation by combining the region's own state with edge-conditioned messages from neighboring regions. Therefore, after L message-passing layers, $\mathbf{z}_c$ contains load and boundary-link context from the L -hop region-graph neighborhood of region c . The edge readout further combines $\mathbf{z}_c$, $\mathbf{z}_{c'}$, and $\mathbf{e}_{c \rightarrow c'}$ to produce a direction-specific representation for action $c \rightarrow c'$. This design is useful for inter-region routing because congestion depends not only on the current region, but also on neighboring-region load states and the boundary ISLs used by different outgoing directions.

C. POMDP Design

We model the upper-layer planning process for one task as a partially observable Markov decision process (POMDP), denoted by $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \mathcal{O}, \gamma)$.

1) *State S*: At planning step h , let c_h denote the current region. The latent state is

$$s_h = (G^{\text{reg}}, L(t), A^{\text{link}}(t), c_h, c_k^{\text{dst}}, V_h, \tau_h, \tilde{d}_h^{\text{man}}), \quad (27)$$

where $L(t)$ and $A^{\text{link}}(t)$ denote the load and link-availability snapshots at slot t_k , c_k^{dst} is the destination region, V_h is the visited-region set, τ_h is the segment-start time proxy, and $\tilde{d}_h^{\text{man}}$ is the Manhattan distance from c_h to the destination region c_k^{dst} .

2) *Observation O*: The local observation at planning step h is denoted by $o_h \in \mathcal{O}$, and only the current region accesses the local observation

$$o_h = \left[\mathbf{z}_{c_h}; \text{concat} \left(\left\{ \mathbf{z}_{c_h \rightarrow c_h^{(a)}}^e \right\}_{a \in \mathcal{A}} \right); \tilde{d}_h^{\text{man}}; \mathbf{1}_{V_h}; \mathbf{m}_h \right], \quad (28)$$

where $c_h^{(a)}$ denotes the neighboring region reached from c_h by action a , $\mathbf{1}_{V_h}$ is the visited-region mask induced by V_h , and $\mathbf{m}_h = [m_{h,a}]_{a \in \mathcal{A}}$ is the feasible-action mask.

3) *Action Space A*: An inter-region action is defined as $a_h \in \mathcal{A} = \{\text{Up, Right, Down, Left}\}$. These four actions specify only the next region direction and do not determine a concrete boundary ISL pair or a satellite-level next hop. For any $a \in \mathcal{A}$, let $c_h^{(a)}$ denote the neighboring region reached from the current region c_h by action a . To ensure transition-level validity, we introduce a feasible-action mask $\mathbf{m}_h = [m_{h,a}]_{a \in \mathcal{A}}$ during policy sampling and define the feasible action set as

$$\mathcal{A}_h = \{a \in \mathcal{A} \mid m_{h,a} = 1\}. \quad (29)$$

The mask removes three types of moves: backtracking to c_{h-1} , revisiting a region already in V_h , and directions that are currently unavailable for cross-region transition. If $\mathcal{A}_h = \emptyset$, the episode terminates as failure.

4) *Transition and Termination P*: After selecting a_h , the next region is $c_{h+1} = c_h^{(a_h)}$. The environment then attempts to instantiate the corresponding satellite-level segment P_h^{seg} on the planning snapshot at slot t_k . This planning-snapshot feasibility check is the initial construction step; the segment is later revalidated at its actual execution-start slot, where capacity blocking or runtime link failure may trigger local repair or task failure. Since the transition kernel is high-dimensional and time-varying, we adopt a model-free DRL approach.

5) *Reward r*: Since one episode corresponds to the route-setup process of a single task, the delay-related cost for a planning step h that successfully instantiates a satellite-level segment is defined as

$$c_h^{\text{delay}} = \sum_{q=0}^{Q_{k,h}-1} \left(\tau_{v_{k,h,q} v_{k,h,q+1}}^{\text{prop}}(t_{k,h,q}) + \tau_{k,h,q}^{\text{tx}} \right), \quad (30)$$

where $P_h^{\text{seg}} = [v_{k,h,0}, v_{k,h,1}, \dots, v_{k,h,Q_{k,h}}]$ is the ordered satellite sequence of the segment instantiated at planning step h and $Q_{k,h}$ is the hop count of segment P_h^{seg} . The balance-related cost is

$$c_h^{\text{bal}} = w_1 \cdot \bar{\rho}_{c_h, a_h}^{\text{out}} + w_2 \cdot \bar{\rho}_{c_{h+1}}^{\text{in}}, \quad (31)$$

where $\bar{\rho}_{c_h, a_h}^{\text{out}}$ is the mean utilization of boundary ISLs leaving c_h along a_h , $\bar{\rho}_{c_{h+1}}^{\text{in}}$ is the mean utilization of internal ISLs in c_{h+1} , and $w_1, w_2 \geq 0$ are normalization weights. The running reward is

$$r_h^{\text{run}} = - \left(w_{\text{delay}} \cdot c_h^{\text{delay}} + w_{\text{bal}} \cdot c_h^{\text{bal}} \right). \quad (32)$$

This reward serves as a practical step-wise training objective whose preference direction is consistent with that of the long-term objective in Section III, while remaining computable from local segment-level quantities during route setup. Let $I_{\text{succ}} \in \{0, 1\}$ denote the success indicator of the whole trajectory, $T_{\text{ep}} \leq H_{\text{max}}^{\text{reg}}$ denote the realized number of inter-region route-setup steps, and $I_h^{\text{seg}} \in \{0, 1\}$ denote whether planning step h successfully instantiates the satellite-level segment P_h^{seg} . The implemented reward for PPO is defined as

$$r_h = I_h^{\text{seg}} r_h^{\text{run}} + I_{\text{succ}} \frac{R_{\text{succ}}}{T_{\text{ep}}} - (1 - I_{\text{succ}}) \mathbb{I}\{h = T_{\text{ep}} - 1\} R_{\text{fail}}, \quad h = 0, \dots, T_{\text{ep}} - 1. \quad (33)$$

D. Proximal Policy Optimization

On the POMDP $\mathcal{M}$, we adopt PPO-Clip [35] with generalized advantage estimation (GAE) [36] to learn the parameter-sharing inter-region policy $\pi_{\theta}(a_h \mid o_h)$ and value function $V_{\psi}(o_h)$. Let (ϕ, θ, ψ) denote the shared encoder, policy-head, and value-head parameters. During deployment, for each newly generated task, regions first participate in a finite number of synchronized local-neighbor message-passing rounds on the region graph to generate slot-level local embeddings from the current planning snapshot. At each planning step, only the currently active region invokes the shared policy and uses its local observation, while the environment instantiates the selected segment under the current execution constraints. The same slot-level embeddings are reused for the sequential region-hop decisions of this task's route setup. The DRL decision is made from local observations with shared policy parameters after region-level neighbor synchronization.

The policy head uses the action-wise feature

$$\mathbf{X}_h^{\pi}(a) = [\mathbf{z}_{c_h}, \mathbf{z}_{c_h \rightarrow c_h^{(a)}}^e, \tilde{d}_h^{\text{man}}, m_{h,a}], \quad (34)$$

where $c_h^{(a)}$ is the neighboring region reached by action a . These action-wise features are concatenated to form $\mathbf{X}_h^{\pi}$. The value head uses

$$\mathbf{X}_h^V = [\mathbf{z}_{c_h}, \text{mean}_{a \in \mathcal{A}_h}(\mathbf{z}_{c_h \rightarrow c_h^{(a)}}^e), \tilde{d}_h^{\text{man}}, \mathbf{m}_h]. \quad (35)$$

The outputs are

$$\mathbf{z}_h = f_{\theta}(\mathbf{X}_h^{\pi}; \phi), \quad V_{\psi}(o_h) = g_{\psi}(\mathbf{X}_h^V; \phi). \quad (36)$$

We apply feasibility masking

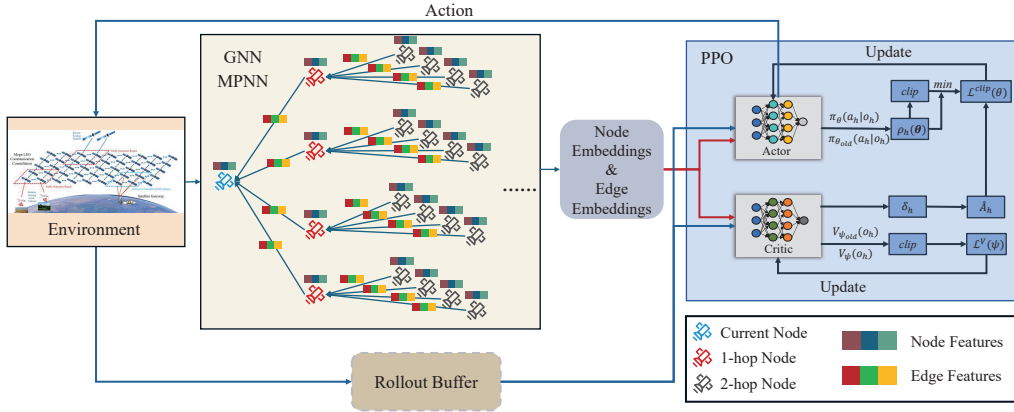


Fig. 3: The framework of GNN-assisted PPO.

$$\tilde{z}_{h,a} = z_{h,a} + (1 - m_{h,a})(-M), \quad M \gg 1, \quad (37)$$

so that probability mass is assigned only to feasible actions, and define

$$\pi_{\theta}(a_h | o_h) = \text{softmax}(\tilde{\mathbf{z}}_h)_{a_h}. \quad (38)$$

Let $d_h \in \{0, 1\}$ denote the episode-termination indicator after step h .

For an episode trajectory $\{(o_h, a_h, r_h, d_h)\}_{h=0}^{T_{\text{ep}}-1}$ with $T_{\text{ep}} \leq H_{\text{max}}^{\text{reg}}$, the TD residual, GAE advantage, and return target are

$$\delta_h = r_h + \gamma(1 - d_h) \cdot V_{\psi_{\text{old}}}(o_{h+1}) - V_{\psi_{\text{old}}}(o_h), \quad (39)$$

$$\hat{A}_h = \sum_{l=0}^{T_{\text{ep}}-1-h} (\gamma \lambda_{\text{GAE}})^l \left(\prod_{j=0}^{l-1} (1 - d_{h+j}) \right) \delta_{h+l}, \quad (40)$$

$$\hat{R}_h = \hat{A}_h + V_{\psi_{\text{old}}}(o_h). \quad (41)$$

The advantage is normalized within each PPO update:

$$\hat{A}_h \leftarrow \frac{\hat{A}_h - \mu_{\hat{A}}}{\sigma_{\hat{A}} + \epsilon_{\text{adv}}}, \quad (42)$$

where $\mu_{\hat{A}}$ and $\sigma_{\hat{A}}$ are the mean and standard deviation, and $\epsilon_{\text{adv}} > 0$ is a small constant for numerical stability.

The PPO ratio is

$$\rho_h(\theta) = \frac{\pi_{\theta}(a_h | o_h)}{\pi_{\theta_{\text{old}}}(a_h | o_h)}. \quad (43)$$

The clipped PPO objective is

$$\mathcal{L}^{\text{clip}}(\theta) = \mathbb{E}_h \left[\min \left(\rho_h(\theta) \hat{A}_h, \text{clip}(\rho_h(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_h \right) \right], \quad (44)$$

where $\epsilon > 0$ is the clipping threshold and we monitor

$$\widehat{\text{KL}} = \mathbb{E}_h [\log \pi_{\theta_{\text{old}}}(a_h | o_h) - \log \pi_{\theta}(a_h | o_h)]. \quad (45)$$

For value regression, we use

$$V_{\psi}^{\text{clip}}(o_h) = V_{\psi_{\text{old}}}(o_h) + \text{clip}(V_{\psi}(o_h) - V_{\psi_{\text{old}}}(o_h), -\epsilon, \epsilon), \quad (46)$$

and let the value loss take the larger one of the unclipped and clipped losses:

$$\mathcal{L}^V(\psi) = \mathbb{E}_h \left[\max \left(\ell(V_{\psi}(o_h) - \hat{R}_h), \ell(V_{\psi}^{\text{clip}}(o_h) - \hat{R}_h) \right) \right], \quad (47)$$

where $\ell(\cdot)$ denotes the squared-error loss. To encourage exploration, an entropy regularizer is introduced:

$$\mathcal{H}(\pi_{\theta}(\cdot | o_h)) = - \sum_{a \in \mathcal{A}_h} \pi_{\theta}(a | o_h) \log \pi_{\theta}(a | o_h). \quad (48)$$

To stabilize early-stage training under sparse successful trajectories, we employ a heuristic teacher-guided warm start based on a cost-aware Dijkstra search on the region graph under the current resource snapshot, yielding a reference region path $\mathbf{c}^T = [c_0, c_1, \dots, c_{K_T}]$. The teacher cost is

$$\text{cost}(c \rightarrow n) = w_{\text{delay}} \tilde{d}_{n \rightarrow c_k^{\text{dst}}}^{\text{man}} + w_{\text{bal}} (w_1 \bar{\rho}_{c,d(c,n)}^{\text{out}} + w_2 \bar{\rho}_n^{\text{in}}), \quad (49)$$

where $\tilde{d}_{n \rightarrow c_k^{\text{dst}}}^{\text{man}}$ is the Manhattan distance from region n to the destination region c_k^{dst} , and $d(c, n)$ is the direction from c to n . The teacher is intentionally used as a heuristic reachability guide rather than an optimal routing oracle. It is used to provide early successful trajectories when PPO rarely reaches the destination by itself. The final policy is not trained to imitate this heuristic. The delay and load-balance preferences are optimized through the PPO reward and update in Eq. (33) and Eq. (52) after reachability is learned. At the start of each episode, we sample $u \sim \text{Bernoulli}(\epsilon_T)$. If the teacher action a_h^T is feasible, a rollout-only logit bonus is added:

$$\tilde{z}_{h,a} \leftarrow \tilde{z}_{h,a} + \beta_T \mathbb{I}[a = a_h^T], \quad (50)$$

where $\beta_T > 0$ is the teacher bonus. This bonus is used only for rollout sampling and is removed during optimization.

In our implementation, policy forward uses raw masked logits without teacher-logit injection during optimization. Moreover, the PPO policy loss, likelihood-ratio clipping, empirical KL, and clip fraction are computed on samples from non-teacher episodes only. Teacher-guided samples are excluded from the PPO policy loss, so Eq. (43)–Eq. (44) are evaluated on the non-teacher data stream.

The teacher provides additional supervision through the behavior cloning (BC) loss

$$\mathcal{L}^{\text{BC}} = -\frac{1}{\sum_h \nu_h} \sum_h \nu_h \log \pi_\theta(a_h^T | o_h), \quad (51)$$

and $\mathcal{L}^{\text{BC}} = 0$ when no valid teacher labels exist in the mini-batch.

Combining the above terms, we minimize the total loss:

$$\begin{aligned} \min_{\phi, \theta, \psi} \mathcal{J} = & -\mathcal{L}^{\text{clip}}(\theta) + c_v \mathcal{L}^V(\psi) \\ & - c_{\text{ent}} \mathbb{E}_h \left[\mathcal{H}(\pi_\theta(\cdot | o_h)) \right] + c_{\text{bc}} \mathcal{L}^{\text{BC}}, \end{aligned} \quad (52)$$

where $\mathcal{L}^{\text{clip}}$ and the entropy term are computed on the non-teacher subset $\mathcal{M}_{\text{ppo}}$, $\mathcal{L}^{\text{BC}}$ on the teacher-valid subset $\mathcal{M}_{\text{bc}}$, and $\mathcal{L}^V$ on the full mini-batch.

Algorithm 1 GNN-Assisted PPO

```

1: Initialize encoder  $\phi$ , policy head  $\theta$ , value head  $\psi$ , actor optimizer on  $(\phi, \theta)$ , critic
   optimizer on  $\psi$ , and schedulers  $(\epsilon_T, c_{\text{bc}}, c_{\text{ent}})$ .
2: for iteration  $i = 1$  to  $I_{\text{max}}$  do
3:   Collect rollouts in buffer  $\mathcal{B}$ .
4:   for each episode do
5:     Sample teacher flag once:  $u \sim \text{Bernoulli}(\epsilon_T)$ .
6:     Build local region features by (19) and (20).
7:     Perform  $L$  synchronized local-neighbor message-passing rounds using (21)–
       (24).
8:     if  $u = 1$  then
9:       Run cost-aware Dijkstra using (49) to obtain the teacher path.
10:    end if
11:    for planning step  $h = 0$  to  $H_{\text{max}}^{\text{reg}} - 1$  do
12:      Build  $\mathbf{X}_h^a(a)$ ,  $\mathbf{X}_h^V$ ; compute logits/value via (36).
13:      Apply feasibility masking via (37).
14:      if  $u = 1$  and teacher action valid then
15:        Add teacher logit bonus via (50).
16:      end if
17:      Sample action from the masked policy via (38).
18:      Execute the selected inter-region transition.
19:      if  $d_h = 1$  then
20:        break
21:      end if
22:    end for
23:  end for
24:  Compute  $\hat{A}_h, \hat{R}_h$  by (39)–(41); normalize by (42).
25:  for PPO epoch  $k = 1$  to  $K_{\text{ppo}}$  do
26:    for mini-batch  $\mathcal{M} \subset \mathcal{B}$  do
27:      Recompute raw masked logits and values on  $\mathcal{M}$ .
28:      Split  $\mathcal{M}$  into  $\mathcal{M}_{\text{ppo}}$  and  $\mathcal{M}_{\text{bc}}$ .
29:      Compute PPO-only policy loss  $\mathcal{L}^{\text{clip}}$  by (44) on  $\mathcal{M}_{\text{ppo}}$ .
30:      Compute value loss  $\mathcal{L}^V$  by (46) and (47) on  $\mathcal{M}$ .
31:      Compute entropy by (48) on  $\mathcal{M}_{\text{ppo}}$  and BC by (51) on  $\mathcal{M}_{\text{bc}}$ .
32:      Update  $(\phi, \theta)$  using the actor loss in (52) with global-norm clipping.
33:      Update  $\psi$  using  $\mathcal{L}^V$  with global-norm clipping.
34:    end for
35:  end for
36:  Monitor PPO-only KL by (45); update  $\epsilon_T$  and  $c_{\text{bc}}$ .
37: end for
38: Return  $(\phi, \theta, \psi)$ .
```

After each PPO update, we compute the success rate of non-teacher episodes over the current update window and update $(\epsilon_T, c_{\text{bc}})$ by a threshold rule: when the rate exceeds a threshold, both are decreased by fixed steps. The entropy coefficient is annealed once per update by $c_{\text{ent}} \leftarrow \max(c_{\text{ent}}^{\text{min}}, \alpha_{\text{ent}} c_{\text{ent}})$. In optimization, we use two optimizers with a shared encoder. We first compute gradients of the total objective in Eq. (52), then apply them by parameter groups: the actor optimizer updates (ϕ, θ) , while the critic optimizer updates ψ only.

E. Intra-Region Path Construction

At each successful upper-layer transition ($c_h \rightarrow c_{h+1}$), the environment constructs one executable satellite-level segment on the planning snapshot at slot t_k . Let $\mathcal{G}_{c_h} = (\mathcal{V}_{c_h}, \mathcal{E}_{c_h})$ denote the intra-region topology of the current region c_h , and let s_h^{in} and s_h^{out} denote the entry and exit satellites of the segment to be instantiated. For the selected region transition ($c_h \rightarrow c_{h+1}$) at planning step h , let $\mathcal{B}_{c_h \rightarrow c_{h+1}}(t_k)$ denote the set of available boundary ISLs between the two regions on the planning snapshot. If $\mathcal{B}_{c_h \rightarrow c_{h+1}}(t_k) = \emptyset$, the current route-setup episode terminates as failure. Otherwise, we select the boundary pair $(u^*, v^*) \in \mathcal{B}_{c_h \rightarrow c_{h+1}}(t_k)$ by scanning the available boundary ISLs using actual ISL load information within the selected region pair and choosing the pair with the smallest snapshot cost. We then set $s_h^{\text{out}} = u^*$ as the exit satellite of region c_h and $s_{h+1}^{\text{in}} = v^*$ as the entry satellite of region c_{h+1} .

We formulate intra-region path construction as a minimum-cost path search from s_h^{in} to s_h^{out} on $\mathcal{G}_{c_h}$. Since $\mathcal{G}_{c_h}$ is small, we adopt a load-aware Dijkstra algorithm whose edge cost combines hop count and link-load terms to bypass congested links while controlling path-length growth. For any local edge $(u, v) \in \mathcal{E}_{c_h}$, ρ_{uv} denotes the link-occupancy ratio corresponding to $\rho_{ij}(t_k)$ in Section III-B. Let $\rho_{\min} = \min_{(u,v) \in \mathcal{E}_{c_h}} \rho_{uv}$ and $\rho_{\max} = \max_{(u,v) \in \mathcal{E}_{c_h}} \rho_{uv}$. We normalize the link occupancy by

$$\tilde{\rho}_{uv} = \begin{cases} \frac{\rho_{uv} - \rho_{\min}}{\rho_{\max} - \rho_{\min}}, & \rho_{\max} > \rho_{\min}, \\ 0, & \rho_{\max} = \rho_{\min}. \end{cases} \quad (53)$$

We then define the local edge weight used by the intra-region Dijkstra search as a linear combination of a hop term and a load term:

$$w_{uv} = w_{\text{hop}} + \beta_{\text{load}} \tilde{\rho}_{uv}, \quad (54)$$

where w_{hop} is the per-hop cost weight and β_{load} is the nonnegative load-cost coefficient shared by the local links. Accordingly, the total cost of the intra-region segment P_h^{seg} is

$$\text{Cost}(P_h^{\text{seg}}) = \sum_{(u,v) \in P_h^{\text{seg}}} w_{uv}. \quad (55)$$

With this cost design, higher link load leads to a larger edge cost, so the planner tends to select edges with lower load. If no feasible path exists under the selected boundary pair, the current route-setup episode terminates as failure.

Otherwise, the resulting path is used as the executable intra-region segment P_h^{seg} for the current planning step.

At execution time, the planned segment is revalidated using the latest link availability and reservation state. Both link disruption and capacity blocking can trigger one local repair attempt. The region reruns the load-aware Dijkstra algorithm once on the updated local topology and the latest load snapshot to obtain a new feasible segment. If a boundary ISL becomes unavailable or capacity-blocked, another feasible boundary pair between the same two regions is selected. This repair step uses the concrete boundary-link and intra-region link states of the selected boundary. If the boundary pair is successfully changed, the entry satellite of the next region will be altered, automatically triggering another intra-region replanning of the next region. If local repair and boundary-pair reselection fail, the current task is regarded as failed.

F. Complexity Analysis

Assume that, in each planning episode, the region graph contains $N_{\text{reg}} = |\mathcal{V}^{\text{reg}}|$ region nodes and $M_{\text{reg}} = |\mathcal{E}^{\text{reg}}|$ directed edges. Let d_n and d_e denote the node-feature and edge-feature dimensions, respectively, H the GNN hidden dimension, H_e the edge-embedding dimension, d_r the edge-readout hidden dimension, L the number of message-passing layers, $H_{\text{max}}^{\text{reg}}$ the maximum number of inter-region route-setup steps, and d_p the hidden dimension of the PPO policy and value heads. For intra-region path construction, let $\mathcal{G}_{c_h} = (\mathcal{V}_{c_h}, \mathcal{E}_{c_h})$ denote the current region subgraph. Since the region graph is encoded once in each planning episode, the overall online complexity for a single task is

$$\begin{aligned} C_{\text{online}} = & \mathcal{O}\left(N_{\text{reg}}d_nH + LM_{\text{reg}}H(2H + d_e) + H_{\text{max}}^{\text{reg}}|\mathcal{A}|d_p^2\right. \\ & + M_{\text{reg}}((2H + d_e)d_r + d_rH_e) \\ & \left. + R(|\mathcal{V}_{c_h}| + |\mathcal{E}_{c_h}|) \log |\mathcal{V}_{c_h}|\right). \end{aligned} \quad (56)$$

This equation describes the nominal single-task online complexity when each traversed region triggers one local shortest-path computation on $\mathcal{G}_{c_h}$ to construct the corresponding segment P_h^{seg} .

V. SIMULATION RESULTS

This section presents a simulation analysis of the proposed hierarchical routing strategy. First, the simulation parameters are introduced. Next, a parametric analysis is performed to illustrate the impact of PPO hyperparameters and to verify the effectiveness of the teacher-guided warm start mechanism through ablation tests. Finally, the proposed method is compared with benchmark algorithms under both stable link conditions and runtime ISL failures.

A. Simulation Setup

We consider a mega LEO constellation constructed from real TLE data. In the simulation, the constellation is instantiated as a Walker-delta-like LEO shell with 60 orbital planes and 60 satellites per plane, yielding $N = 3600$. The Walker

structure defines the orbital-plane index, the in-plane satellite index, the regular candidate ISL neighborhood, and wrap-around connections, while TLE/SGP4 propagation provides time-varying satellite positions, link distances, and gateway visibility. A single gateway station provides S2G connectivity. To emulate spatially localized bursty traffic, one task is generated per slot. During each refresh window, the source and gateway-visible egress satellite are repeatedly sampled from predefined patches. The destination egress satellite is selected from the satellites visible to the gateway station, while the source patch is randomly chosen so that it does not overlap with the destination patch. The patch side length is set to $L_{\text{patch}} = 8$ satellites, and the patch is refreshed every $I_{\text{refresh}} = 600$ time slots to represent hotspot movement. For each generated task, each route is first planned on the slot snapshot and then checked during execution.

The main system parameters, together with the main GNN and PPO hyperparameters, are summarized in Table II. For fair comparison, all algorithms use the same task-generation process, task-size setting, reservation-based resource-allocation, execution-time capacity check, and runtime ISL failure setting. The global execution baselines output a full satellite-level path and are checked by the global execution layer without hierarchical segment-level local repair. The hierarchical execution baselines use the same region partition, lower-layer path construction, execution-time revalidation, and local repair mechanism as the proposed method. Failed tasks are counted with a fixed 30s penalty in the penalized average delay. The baseline implementations are specified as follows:

1) *Global centralized (GC)*: This baseline is a centralized satellite-level Dijkstra method on the full 3600-node physical graph, inspired by [37]. GC uses a load-aware shortest-hop Dijkstra mode. The load of each feasible physical ISL is defined as the cost, and currently blocked physical ISLs are removed from the search. After the full satellite path is obtained, it is checked by the same time-indexed reservation and execution-time capacity mechanism used by the other algorithms. If the planned global path contains a disrupted or capacity-blocked link during execution, the task fails because GC does not include the hierarchical segment repair or boundary-pair reselection used by the hierarchical methods.

2) *Global GNN-assisted PPO (GGAP)*: This baseline, inspired by [38], evaluates the effect of removing the proposed regional hierarchy while keeping a GNN-assisted PPO planner. GGAP performs next-hop routing directly on the full 3600-node satellite graph. For each task, the physical satellite graph is constructed from the current slot snapshot, where each satellite has four directed candidate ISL actions corresponding to the top, right, bottom, and left physical neighbors with wrap-around connectivity. For each satellite node i , GGAP uses a 7-dimensional node feature vector that includes normalized column and row distances from satellite i to the destination satellite, mean load ratio of the four outgoing candidate ISLs of satellite i , minimum headroom ratio among these outgoing candidate ISLs, normalized satellite coordinates, and a binary indicator of whether satellite i is the current task destination. For each directed physical ISL ($i \rightarrow j$), GGAP uses a 3-dimensional edge feature, including load ratio of the directed

TABLE II: SIMULATION PARAMETERS

Parameter	Value
Orbital inclination i	57
Number of orbital planes N_L	60
Satellites per plane M_L	60
Phase factor F	23
Number of satellites N	3600
Minimum elevation angle $\theta_{\min}$	10.0°
Time slot length Δt	0.1 s
Total simulation slots T_{sim}	30000
ISL nominal rate C	10000 Mbps
Background occupancy ratio ρ_{ij}^{bg}	0.001
Per-task occupancy ratio α_0	0.2
Blocking threshold ratio ρ_{blk}	0.9
Runtime-failure per-link probability p_{lf}	0.03
Runtime-failure trigger period P_{lf}	1 slot
Runtime-failure duration D_{lf}	20 slots
Region side length S_c	5
Number of regions N_{reg}	144
Regions per row $N_{\text{reg},\text{row}}$	12
Max inter-region route-setup steps $H_{\text{max}}^{\text{reg}}$	24
GNN layers L	3
GNN hidden dimension H	64
Edge embedding dimension H_e	64
Edge readout hidden dimension d_r	128
PPO head hidden dimension d_p	128
Discount factor γ	0.99
GAE parameter λ_{GAE}	0.95
PPO clipping threshold ϵ	0.2
Value loss coefficient c_v	0.5
Initial teacher intervention probability ϵ_T	0.35
Teacher logit bonus β_T	4
Warm-start success-rate threshold ζ_{ws}	0.3
Teacher-probability decay step $\Delta\epsilon_T$	0.05
Mini-batch size B_{mb}	512
PPO epochs per update K_{ppo}	3
Failure penalty R_{fail}	5.0
Terminal bonus R_{succ}	3.0

ISL, headroom ratio, and a binary blocked-link indicator. GGAP uses the same edge-conditioned GraphSAGE form and the same main PPO hyperparameters as the proposed method. At each planning step, the current satellite scores its four candidate physical next-hop actions. The action-wise policy input concatenates the current satellite embedding, the candidate-neighbor embedding, the corresponding directed-edge embedding, goal-related scalars, direction-risk terms, and the graph readout. The feasible-action mask removes unavailable or capacity-blocked physical ISLs. Previously visited satellites are not hard-masked, but revisits are penalized in the reward. The planned satellite-level path is finally checked by the execution-time capacity validation. Hierarchical segment-level local repair is not applied in this baseline.

3) *Hierarchical Without GNN (No_GNN)*: This baseline, inspired by baseline algorithm selection in our previous work [39], is designed to isolate the contribution of GNN-based representation learning. It keeps the same hierarchical POMDP, four-direction region action space, feasible-action mask, reward design, PPO training hyperparameters, lower-layer path construction, execution-time capacity check, and local-repair mechanism as the proposed method, while removing the GNN encoder and the direction-specific edge readout. No_GNN directly feeds handcrafted region-level features into PPO. Specifically, it uses the same 22-dimensional local regional context as the proposed method, as described earlier. For each

candidate region direction, the actor input is formed by directly concatenating the current-region feature, candidate-neighbor region feature, directed edge feature, goal-related scalars, and direction-risk terms. Therefore, No_GNN keeps the same compact input categories and the same PPO training setting as the proposed method, but removes topology-conditioned GNN propagation and learned action-specific edge readout.

4) *Heuristic hierarchical (HH)*: This baseline, inspired by [40], evaluates whether the hierarchical decomposition alone is sufficient without a learned inter-region policy. HH keeps the same region partition, lower-layer path construction, execution-time revalidation, and local repair mechanism as the proposed method, but it does not construct or train any GNN/PPO model. At the upper layer, HH performs deterministic hop-by-hop region selection. At each step, it enumerates the four neighboring regions and selects the direction with the smallest normalized region-level Manhattan distance to the destination region. After the heuristic region path is selected, HH uses the exact same lower-layer segment-instantiation mechanism as the proposed method. For each selected inter-region transition, the concrete boundary ISL pair is chosen by scanning candidate boundary pairs and using the same boundary-pair cost form as the hierarchical execution module, with distance and normalized boundary-load terms. The corresponding intra-region segment is then constructed by the shared load-aware shortest-path routine. Therefore, HH is a deterministic hierarchical heuristic baseline: it preserves the coarse-to-fine execution and local repair structure, but removes both GNN-based representation learning and PPO-based inter-region decision making.

B. Parametric Analysis

In this subsection, we conduct experiments to determine the learning rate with the best convergence behavior and to verify the effectiveness of the teacher-guided warm start through ablation experiments.

Since actor updates depend on critic estimates, the critic learning rate is set to twice the actor learning rate. For each setting, we run 10 rounds with 3000 time slots each and smooth the reward curves by exponential moving average (EMA) and sliding-window averaging for readability.

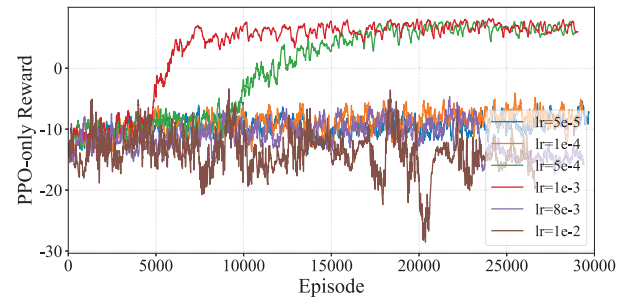


Fig. 4: Comparison of convergence performance under different learning rates.

The results show a clear pattern in Fig. 4. A large learning rate, such as 0.01, causes strong oscillation and unstable training. Moderate values, especially 0.001 and 0.0005, converge

much more reliably, while 0.001 reaches a stable regime faster. In contrast, very small learning rates, such as 0.0001 and 0.00005, make policy improvement too slow. We therefore use 0.001 as the actor learning rate in the main experiments.

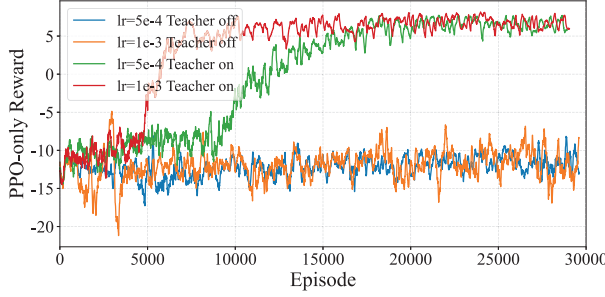


Fig. 5: Ablation test of teacher-guided warm start.

We then evaluate the teacher-guided warm start at the two best learning rates, namely 0.001 and 0.0005. The goal is to test whether teacher guidance improves convergence in the early training stage when successful trajectories are very sparse. As shown in Fig. 5, training with the teacher mechanism converges earlier and remains stable, whereas training without teacher guidance fails to achieve comparable convergence under either learning rate. This indicates that teacher guidance improves early sample quality and thus helps PPO converge faster and more stably.

We also measure the online planning runtime of the proposed method in the above mentioned network. The measured time includes online embedding generation, route setup, and execution-time revalidation, excluding training updates, model loading and checkpoint saving. The proposed method costs 32.38 ms on average, with a median of 31.06 ms and a 95th percentile of 53.58 ms per measured task on an AMD Ryzen 7 5800 CPU.

C. Performance Comparison

This subsection compares all algorithms under two settings with the same task-size sweep. We first study stable link conditions to evaluate the trade-off among task success rate, penalized average delay, and load balance. We then introduce runtime ISL failures on the same setup to examine robustness under link disruptions during execution.

We first introduce HSI as an auxiliary evaluation metric that complements JFI. JFI measures overall load balance, whereas HSI measures load concentration by quantifying how much of the total load is concentrated on a small set of hotspot links. Specifically, HSI is defined as the fraction of total link load carried by the top $\eta\%$ highest-utilization links in each time slot. We set $\eta = 0.01$, and let $k(t) = \max\{1, \lceil \eta \cdot |\mathcal{E}(t)| \rceil\}$ denote the number of top 1% links with the largest utilization values. The set of these links is written as $\mathcal{E}_\eta^{\text{top}}(t) \subseteq \mathcal{E}(t)$. The instantaneous HSI is

$$\text{HSI}(t) = \begin{cases} \frac{\sum_{(i,j) \in \mathcal{E}_\eta^{\text{top}}(t)} \rho_{ij}(t)}{\sum_{(i,j) \in \mathcal{E}(t)} \rho_{ij}(t)}, & \sum_{(i,j) \in \mathcal{E}(t)} \rho_{ij}(t) > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (57)$$

The time-averaged HSI over a run with T_{sim} time slots is

$$\overline{\text{HSI}} = \frac{1}{T_{\text{sim}}} \sum_{t=1}^{T_{\text{sim}}} \text{HSI}(t). \quad (58)$$

A smaller $\overline{\text{HSI}}$ indicates lower load concentration on a small set of heavily utilized links. However, HSI should not be interpreted alone because early task failure can also reduce the amount of carried traffic on hotspot links.

We observe the performance of the algorithms under different nominal task-size levels. Each experimental point is specified by a nominal task-size level B_{nom} . At the point labeled $B_{\text{nom}} = x$ Mb, the task size of each generated task is randomly sampled from the interval $[x - 50, x + 50]$ Mb. For each nominal task-size level, we run the simulation 5 times for each algorithm and present the overall averaged metrics.

1) *Stable link conditions*: Under stable link conditions, links do not fail because of non-geometric events such as APT errors or hardware glitches, that is, $a_{ij}^{\text{otg}}(t) \equiv 1$. Fig. 6 reports the performance under different nominal task-size levels. A larger nominal task size increases the resource demand of each task and makes admission control more restrictive. The proposed method maintains high reachability in all tested scenarios. Averaged over all task-size levels, it achieves a task success rate of 0.9980, a penalized average delay of 9.01 s, and JFI = 0.9853. At $B_{\text{nom}} = 500$ Mb, it still maintains a task success rate of 0.9983 and a penalized average delay of 12.73 s. This shows that the proposed method can preserve routing feasibility and load fairness under large task sizes.

GC gives a strong delay reference in the stable-link case. At $B_{\text{nom}} = 500$ Mb, it achieves a task success rate of 0.9993 and a penalized average delay of 9.40 s. However, its shortest-hop rule leads to poorer load distribution, with JFI = 0.7679 and HSI = 0.3117. No_GNN keeps the same hierarchical execution and repair mechanism as the proposed method, but uses direct engineered-feature inputs instead of GNN message passing and direction-specific edge readout. Its task success rate and penalized average delay are 0.9697 and 15.16 s at $B_{\text{nom}} = 500$ Mb. The gap between No_GNN and the proposed method shows the benefit of topology-conditioned regional embeddings and direction-specific edge representations.

We use HH to further evaluate whether the hierarchical execution structure alone is sufficient by using a distance-based hierarchical heuristic method. HH shares the same path construction and repair setting as the proposed method. At $B_{\text{nom}} = 500$ Mb, HH achieves a task success rate of 0.7070 and a penalized average delay of 16.51 s. This result shows that the hierarchical execution structure needs to be coupled with learned region-level decisions to maintain high reachability under large task sizes. GGAP gives another reference by applying direct satellite-level GNN-PPO routing on the full graph. It also underperforms the proposed method in this

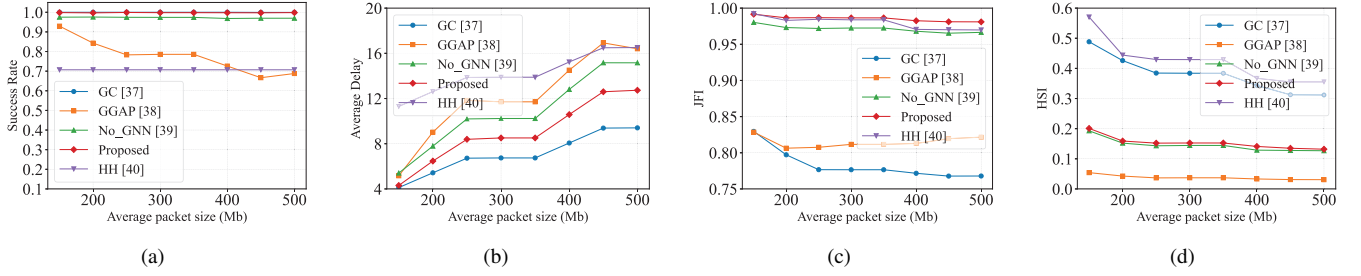


Fig. 6: Algorithm performance under different nominal task-size levels with stable link conditions. (a) Task success rate. (b) Penalized average delay. (c) Jain's fairness index. (d) Hotspot severity index.

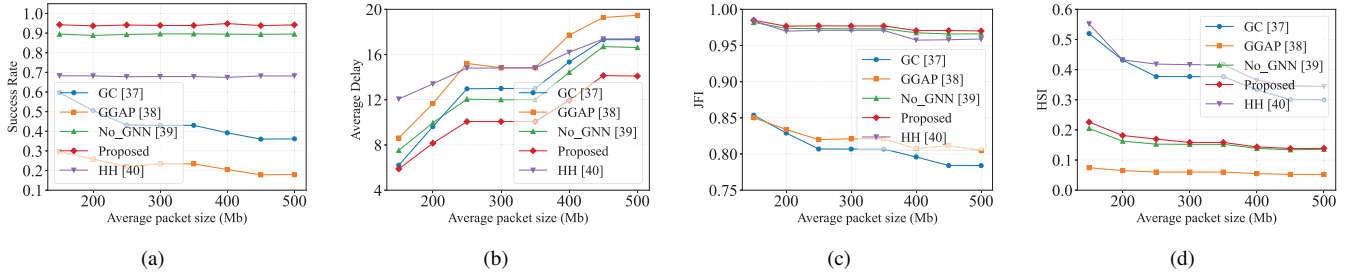


Fig. 7: Algorithm performance under different nominal task-size levels with runtime ISL failures. (a) Task success rate. (b) Penalized average delay. (c) Jain's fairness index. (d) Hotspot severity index.

setting, with task success rate 0.6880 and penalized average delay 16.40 s at $B_{nom} = 500$ Mb. Although GGAP reports a low HSI, this value should not be interpreted as better hotspot control. Its low success rate means that less traffic is carried, which can also reduce the measured hotspot concentration. Overall, under stable link conditions, the proposed method provides a strong trade-off among success rate, penalized average delay, JFI-based load balance, and HSI-based load concentration.

2) *Runtime ISL failures:* We next evaluate the same task-size sweep under runtime ISL failures. In this setting, some ISLs become temporarily unavailable during execution because of non-geometric disruptions such as APT errors or hardware glitches. The runtime-failure generator follows the rule defined in Section III. The same runtime-failure profile and random seed are used for all algorithms. The results are shown in Fig. 7.

Runtime ISL failures make the difference between hierarchical route repair and global path planning clearer. Averaged over all task-size levels, the proposed method achieves a task success rate of 0.9410, a penalized average delay of 10.55 s, and $JFI = 0.9755$. At $B_{nom} = 500$ Mb, it still achieves a task success rate above 0.94 and a penalized average delay of 14.09 s. These results show that the proposed method maintains high reachability and low penalized delay under execution-time link failures.

The global baselines are much more sensitive to runtime ISL failures. At $B_{nom} = 500$ Mb, GC drops to a task success rate of 0.3613, and GGAP drops further to a task success rate of 0.1793. Their penalized delays are also higher than that of the proposed method. Among the hierarchical baselines,

No_GNN remains the closest one, with a task success rate of 0.8953 and a penalized average delay of 16.62 s. HH reaches a task success rate of 0.6820 and a penalized average delay of 17.39 s. These results confirm that hierarchical repair is robust, but GNN-assisted inter-region decisions are still needed for reliable route setup under runtime failures. Overall, the proposed method provides the most balanced performance across the four reported metrics in the runtime-failure setting.

VI. CONCLUSION

This paper studied scalable routing for mLEO satellite constellations under localized hotspot traffic. We proposed a learning-based hierarchical routing framework that decomposes route setup into inter-region route setup and intra-region path construction. At the upper layer, regions generate local embeddings through a finite number of synchronized local-neighbor message-passing rounds on the region graph, and a parameter-sharing GNN-assisted PPO policy selects the inter-region route through sequential decisions based on local observations. At the lower layer, executable satellite paths are constructed by a load-aware shortest-path method, and execution-time revalidation with local repair is used to handle disrupted or capacity-blocked segments. Simulation results obtained using real TLE data and SGP4 propagation under varying task-size levels and runtime ISL failures showed that the proposed method achieves a favorable overall trade-off among task success rate, penalized average delay, JFI-based load balance, and HSI-based load concentration under both stable link conditions and runtime ISL failures.

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